THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 100 J	W = F × s
2	**(d)** 16 J	½mv²
3	**(a)** 100 J	mgh
4	**(c)** 60 W	P = W/t
5	**(d)** 3	load ÷ effort
6	**(c)** 30 N	effort × effort arm = load × load arm
7	**(a)** zero	force ⊥ displacement
8	**(d)** remains constant	PE converted to KE
9	**(d)** load between fulcrum and effort	second class
10	**(d)** 2 : 1	½mv² = ½(4m)v²

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Work = force × displacement in the direction of the force (1); joule (1).	1+1
12	Work done when a force of 1 N displaces a body by 1 m in the direction of the force (2).	2
13	When force is zero, displacement is zero, or force is perpendicular to displacement (2).	2
14	W = 40 × 5 = 200 J (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lifting: force and displacement are upward, so positive (1.5); lowering: force upward, displacement downward, so negative (1.5).	1½+1½
16	Displacement is zero (1.5); the tiredness is due to the use of internal body energy (1.5).	1½+1½
17	Force opposes displacement (1); W = 200 × (−0.15) (1) = −30 J (1).	1+1+1
	OR: W = Fs = ma·s with v² = u² + 2as gives W = ½m(v² − u²) (2); from rest, KE = ½mv² (1).	2+1
18	KE = ½mv²; with 2v it becomes 4 × ½mv² (3).	3
	OR: ½ × 0.2 × 43² (2) = 184.9 J (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) W = F s; 1 J = 1 N × 1 m (2). (ii) Zero: F = 0, s = 0 or F ⊥ s; positive: along; negative: opposite (2). (iii) 150 J (1).	2+2+1
	OR: (i) 1 : 3 : 3 (2). (ii) 50 J (1). (iii) Work increases in proportion to force and to displacement (2).	2+1+2
20	(i) Work = change in energy (1). (ii) W = ½m(v² − u²), KE = ½mv² (2). (iii) ½·15000·v² = 367500·100 ⇒ v = 70 m s⁻¹ (2).	1+2+2
	OR: (i) Work to lift = mg × h (1.5). (ii) Dropping a ball from larger heights makes deeper depressions (2). (iii) KE + PE is constant when no external force acts (1.5).	1½+2+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 150 J	W = F × s
2	**(c)** 180 J	½mv²
3	**(d)** 500 J	mgh
4	**(a)** 50 W	P = W/t
5	**(b)** 4	load ÷ effort
6	**(d)** 20 N	effort × effort arm = load × load arm
7	**(c)** negative	force opposes displacement
8	**(c)** change the direction of the effort	MA = 1
9	**(d)** energy	W = ΔE
10	**(b)** h only (and g)	v = √(2gh)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	When force acts opposite to displacement (1); goalkeeper stopping a ball (1).	1+1
12	Work done on an object equals the change in its energy (2).	2
13	KE = ½mv² (1); PE = mgh (1).	1+1
14	½ × 0.2 × 100 = 10 J (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Force to lift slowly = mg (1); work = mg × h (1); stored as PE = mgh (1).	1+1+1
16	PE = 0.2 × 10 × 10 = 20 J (1.5); ½mv² = 20 ⇒ v = √200 ≈ 14.1 m s⁻¹ (1.5).	1½+1½
17	At height h: PE = mgh, KE = 0 (1); just before ground: PE = 0, KE = ½mv² = mgh since v² = 2gh (2).	1+2
	OR: At extreme positions PE is maximum and KE zero (1); at the lowest point KE is maximum and PE minimum (1); the total is nearly constant, losses are due to friction and air (1).	1+1+1
18	mgh = ½mv² (1); v² = 2 × 10 × 2.5 = 50 (1); v ≈ 7.07 m s⁻¹ (1).	1+1+1
	OR: (i) electrical → light and heat (1); (ii) mechanical → sound (1); (iii) light → chemical (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 100 J (1). (ii) v = √(2 × 10 × 20) = 20 m s⁻¹ (2). (iii) PE = 50 J, so KE = 50 J (1). (iv) Conservation of mechanical energy (1).	1+2+1+1
	OR: (i) A bob released from a point rises to nearly the same height on the other side (2). (ii) Friction at the support and air resistance convert energy to heat (2). (iii) Extreme: PE; mean: KE (1).	2+2+1
20	(i) P = W/t, watt (1). (ii) Work is the energy transferred; power is its rate (2). (iii) W = 18 000 J; P = 900 W (2).	1+2+2
	OR: (i) 40 × 10 × 72.5 = 29 000 J (1). (ii) 29 000 J (1). (iii) PE depends only on the height, not the path (1). (iv) 290 W (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.